

Hypothesis Testing of Variance

Step 1

H_0 : equal to some specific val

H_1 : different (greater or small)

Step 2:

Test statistic

$$\chi^2 = \frac{(n-1)S^2}{\sigma^2}$$

n : sample size

S^2 : Sample var

σ^2 : POP variance

Step 3:

Critical value, $v = n-1$

$$\chi^2_{\alpha} < \chi^2_{1-\alpha/2}$$

Step 4:

~~if χ^2 is within region then reject H_0~~

Right tail : $\chi^2 > \chi^2_{\alpha}(v)$ reject H_0

Left tail : $\chi^2 < \chi^2_{1-\alpha}(v)$ reject H_0

Two tailed : $\chi^2 < \chi^2_{1-\alpha/2}(v)$

OR

$$\chi > \chi^2_{\alpha/2}(v)$$

reject H_0

Practice question:

A factory produces metal rods with a standard deviation of 3 mm in their lengths. A quality control analyst wants to test if the population variance has changed. A random sample of 15 rods is taken, and the sample variance is found to be 10 mm². At a 5% significance level, test whether the population variance is still 9 mm² ($\sigma^2 = 9$) using the chi-square test.

- State the null and alternative hypotheses.
- Determine the test statistic.
- Find the critical value(s) or p-value.
- Conclude whether the population variance has changed.

Given:

$$\sigma^2 = 9$$

$$n = 15$$

$$s^2 = 10$$

$$\alpha = 0.05$$

Step 1:

$$H_0 = \sigma^2 = 9$$

$$H_1 = \sigma^2 \neq 9$$

Step 2:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$$= \frac{(15-1)10}{9} = 15.56$$

Step 3:

$$df = 14$$

$$\alpha = 0.05 \quad \alpha/2 = 0.025$$

$$\chi^2_{0.025}(14) = 5.63$$

$$\chi^2_{0.975}(14) = 26.12$$

Step 4:

$$\chi^2 < 5.63$$

OR

$$\chi^2 > 26.12$$

Step 5:

H_0 Accepted